Representation-based inference

Per-outcome classifier

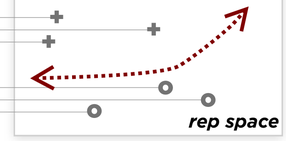
training / tuning

id=644314104

<BOS>, <AGE//age_Q4>, <SEX//male>, <RACE// black_or_african_american>, **...**, <LAB-RES//creatinine_Q1>, <LAB-RES//alt_Q3> **<24 hr>**

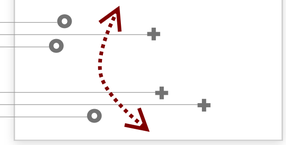


outcome = anemia



...

outcome = pressors



Inference per classifier

held-out

id=678580042

<BOS>, <AGE//age_Q7>, <SEX//female>, <RACE// asian>, **...**, <ASMT// gcs_verbal_Q9>, <VTL// sbp_Q6> **<24 hr>**

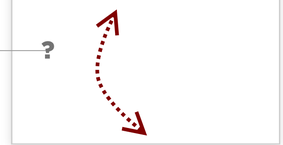


prediction = YES anemia



...

prediction = NO pressors



Generative inference

held-out

id=678580042

<BOS>, <AGE//age_Q7>, <SEX//female>, <RACE// asian>, **...**, <ASMT// gcs_verbal_Q9>, <VTL// sbp_Q6> **<24 hr>**



sample 1 **...**, <LAB-RES//hemoglobin_Q0>, <LAB-RES//wbc_Q7>, **<LABEL//anemia_init>**, ..., <XFR-OUT//ward>, <DSCG//home>,<EOS>

sample 2 **...**, <LAB-RES//pco2_arterial_Q4>, <LAB-RES//lactate_Q2>, **<LABEL//anemia_init>**, ..., <EOS>

sample 3 **...**, <VTL//sbp_Q1>, <VTL//dbp_Q1>, <VTL//temp_c_Q9>, <VTL//respiratory_rate_Q0>, **<LABEL//pressor_init>**, ..., <EOS>

...

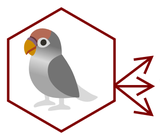
sample *N* **...**,<LAB-RES//lactate_Q2>, **<LABEL//anemia_init>**, ..., **<LABEL//pressor_init>**, ..., <DSCG//assisted_living>, <EOS>

anemia?		pressors?
1		0
1		0
0	...	0
...		...
1		1
count/N		*count/N*

Inference via supervised finetuning

Multiheaded training with custom loss

modified model



Discharge head

TTE head

Next token head

training / tuning

id=644314104

<BOS>, ..., <LAB-ORD//creatinine>, DSCG//home, TTE=5hrs

<BOS>, ..., <LAB-ORD//creatinine>, <VTL//sbp_Q5>, DSCG//home, TTE=4.9hrs

<BOS>, ..., <LAB-ORD//creatinine>, <VTL//sbp_Q5>, <LAB-RES//creatinine_Q1>, DSCG//home, TTE=4.8hrs

DSCG//home, TTE=0hrs

<BOS>, ..., <LAB-ORD//creatinine>, <VTL//sbp_Q5>, <LAB-RES//creatinine_Q1>, ..., <VTL//sbp_Q5>, **<DSCG//home>** <EOS>

Inference

held-out

id=678580042

<BOS>, <AGE//age_Q7>, <SEX//female>, <RACE// asian>, **...**, <ASMT// gcs_verbal_Q9>, <VTL// sbp_Q6>



discharge prediction

TTE prediction